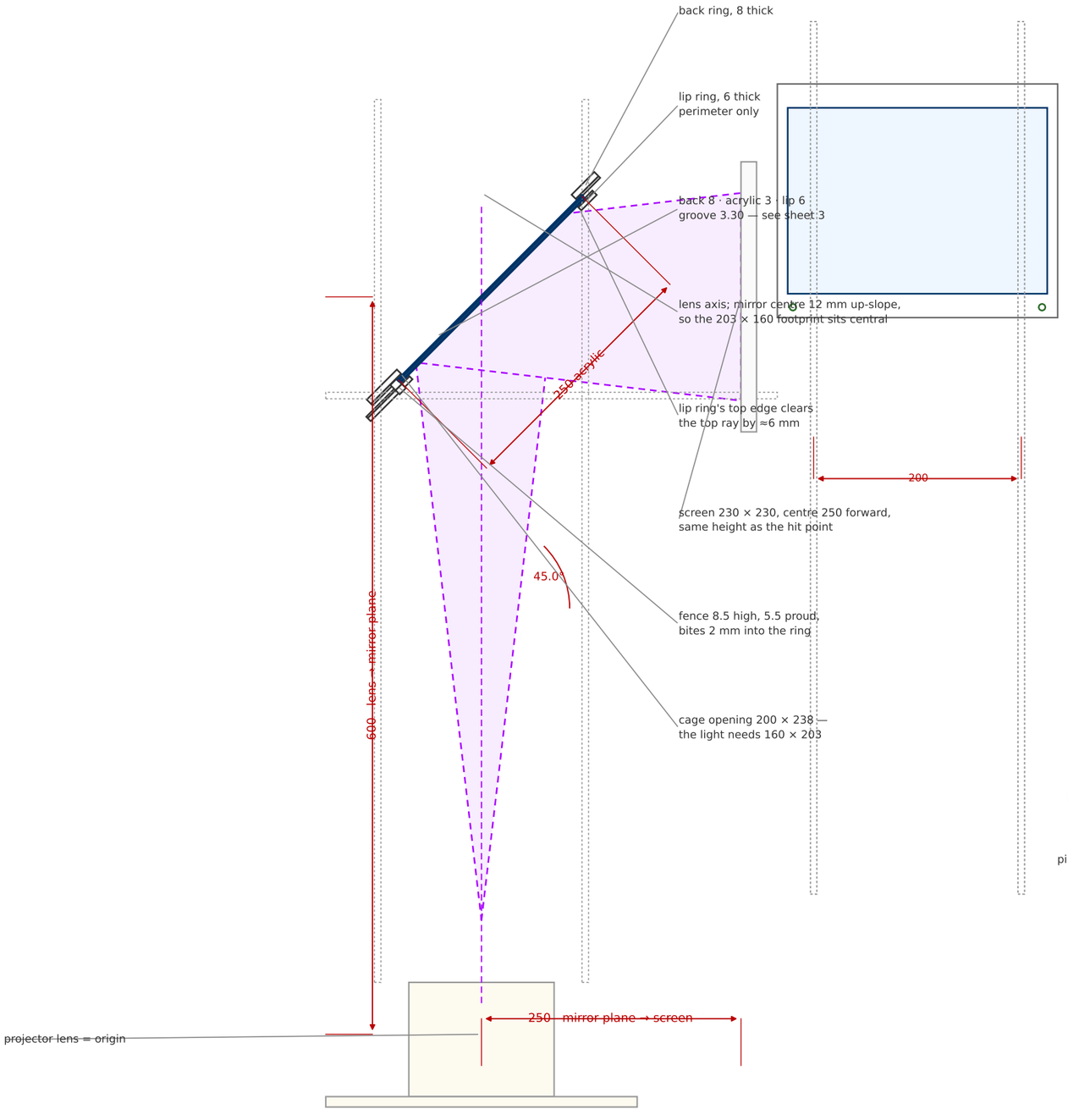
SHEET 1 — ASSEMBLY SECTION

rings hatched · acrylic blue · rods, screen, base grey (behind the cut) · light violet



NOTES

1. Groove 3.30 nominal for 3.00 acrylic. The pads on the screw axes set that width, so tightening the M3s cannot close the groove onto nothing and leave the sheet loose.
2. Second-surface mirror: the coating lies ON the optical plane and the sheet sits in front of it, toward the lens.
3. Pillars Ø6.3 on a 200 × 212 plan pattern; screen-mount bars Ø6.3 at Z 505, Y ±120, through the lower end band.
4. Envelope 225 × 270 × 225 mm. Both rings exceed the SOVOL bed — print on the university machine, or split each ring.
5. Light clearance verified by intersecting the parts with the beam envelopes: both rings miss both the incoming and reflected light.

ORIN RIG — REFLECTOR MOUNT

ASSEMBLY SECTION

cut on the mirror's centre line — world X forward, Z up

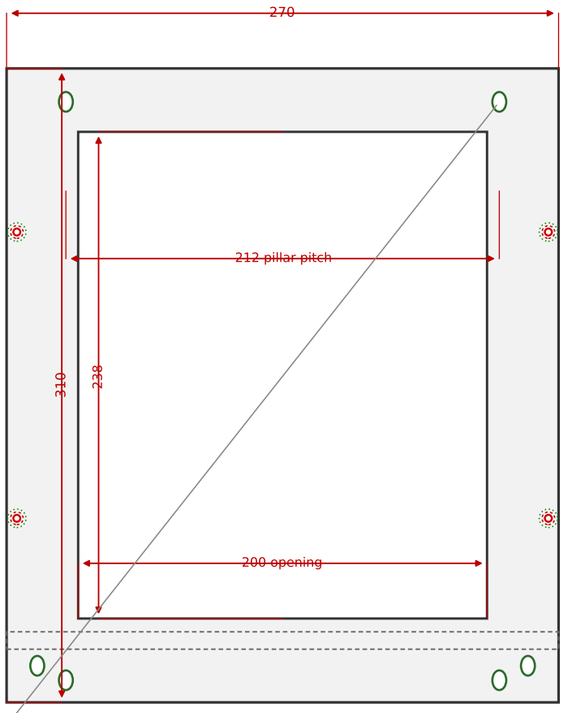
SCALE 1:5 UNITS mm A3 SHEET 1 of 3

PETG, 0.2 mm layers — prototype for a single demonstration

SHEET 2 — PART TRUE VIEWS

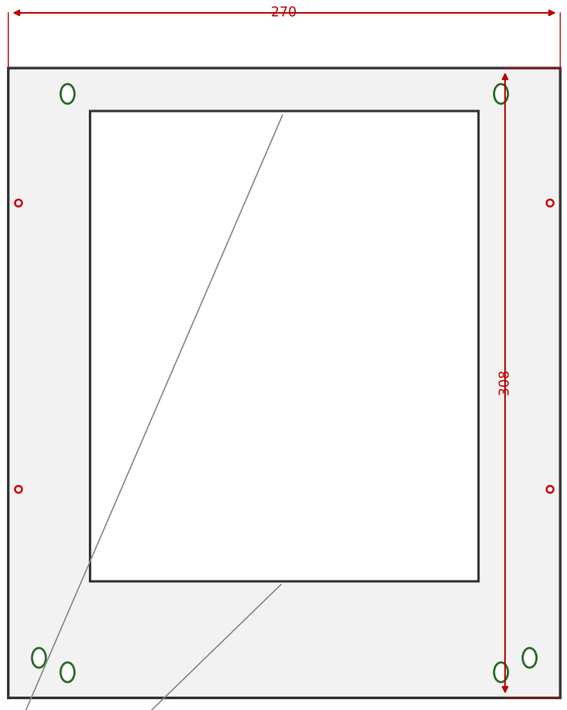
the two printed rings in true shape at 1:3, dimensioned for the print

BACK RING



Ø6.8 pillar bore, world-vertical → 6.8 × 9.6 ellipse
on the face; set screw enters the back face

FRONT LIP RING



190 × 230 opening: the lip touches the sheet only
in the 15 mm margin outside the light

M3 clearance at Y ±130, Z ±70;
pads Ø9 on the back ring

HOLE TABLE		BACK / LIP
4 × Ø6.8	pillar bores, world-vertical (45° to the face)	6.8 / 6.8
2 × Ø6.8	screen-mount bars along X, at Z 505	6.8 / 6.8
4 × Ø3.4	M3 clamp screws at Y ±130, Z ±70	3.4 / 3.4
4 × Ø6.2	hex pockets (AF 5.5), 3 deep, back face	nut / none
4 × Ø4.5	M4 set screws + hex pocket, back face	4.5 / none
4 × Ø9	hard-stop pads, 3.30 long, on the screw axes	none / none

The rings share one 270-wide outline; the lip ring is the shorter of the two and its opening is 10 mm wider in Y and 8 mm shorter in Z, so that it overlaps the sheet by 15 mm all round without shadowing the lit 203 × 160 footprint.

Ellipses read 1.414 × the rod diameter: the rods cross the face at 45°.

ORIN RIG — REFLECTOR MOUNT

PART TRUE VIEWS

looking along the mirror's normal; up-slope is up the page

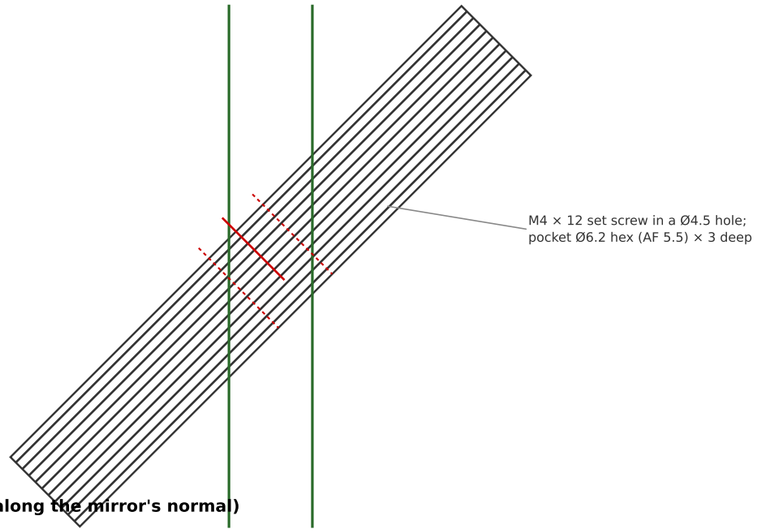
SCALE 1:3 UNITS mm A3 SHEET 2 of 3

PETG, 0.2 mm layers — prototype for a single demonstration

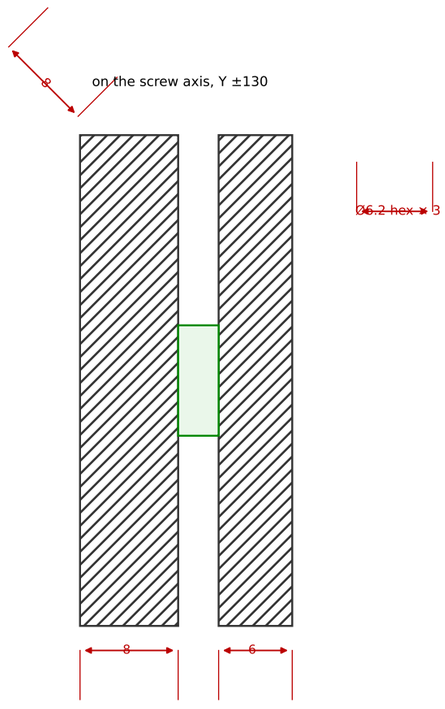
SHEET 3 — DETAILS

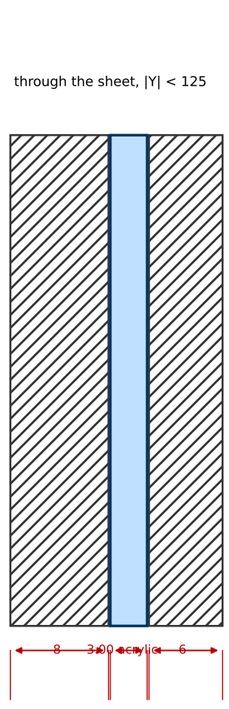
details at 2:1, and the notes that go with the print

A — PILLAR BORE AND SET SCREW (section on the pillar's axis)



B — CLAMP STACK (section along the mirror's normal)





ASSEMBLY NOTES

8 × M3 × 16 + nut clamp the rings; the pad sets the 3.30 groove
4 × M4 × 12 set screw grip the pillars once the 45° is set
1 × mirrored acrylic 250 × 250 × 3, coating on the back face
4 × Ø6.3 × 800 aluminium pillars (1/4")
2 × Ø6.3 × 400 bars — galvanised steel, not aluminium: over 370 mm of cantilever aluminium sags 7.6 mm, steel 2.7 mm. For the 10 mm aluminium tube instead, open the bar bores to Ø10.4.

Fence under the sheet's lower edge: 8.5 high, 5.5 proud, biting 2 mm into the back ring, clearing the acrylic by 0.5.

Print each ring flat, in the mirror's plane, so the bed holds it. The pillar bores then run 45° to the layers: fine at Ø6.8, but expect to ream them.

Light clearance verified: both rings miss both beam envelopes.

ORIN RIG — REFLECTOR MOUNT

DETAILS

pillar bore and set screw · clamp stack

SCALE 2:1 UNITS mm A3 SHEET 3 of 3

PETG, 0.2 mm layers — prototype for a single demonstration